Freight rate prediction: validation and data split

Prediction of the posted rate of 12,000 loads for Nov-Dec 2025, based on 48,000 loads from Jan-Oct 2025.

On loads not seen during training (Sep-Oct), the average error is 1.9% of the rate, about $45 on a load averaging $2,360, and the median error is 1.6%. These figures leave out the 1.4% of loads whose rates are 2 to 5 times above or below the usual rate for that kind of load. Those loads were not found to follow any pattern in the load details, and including them would raise the error to about 4.4% on all Sep-Oct loads.

1. Data split

The model is used to predict later months, so training always uses earlier data and testing uses later data. For this reason, loads are not shuffled between past and future, and the data is divided by period as shown below.

Period	Loads	Use
Jan-Aug 2025	38,477	Training and comparison of setups
Sep-Oct 2025	9,523	Held out to measure performance on unseen months
Nov-Dec 2025	12,000	Loads to be predicted (rates not provided)

For the final predictions, the model is retrained on all 48,000 known loads, which adds Sep-Oct.

2. Validation

Within Jan-Aug, the setups were compared with step-forward checks: training on all months up to a cut-off and predicting the next two months, repeated three times (to Feb for Mar-Apr, to Apr for May-Jun, and to Jun for Jul-Aug). Since each check uses only information available at the time, it reflects how the model is used in practice. The models were then compared on the held-out Sep-Oct months. However, the final choice was made after seeing these results, so the scores are a reasonable but slightly optimistic estimate.

A random mixing of months was avoided for a specific reason. Loads on the same day share the same market conditions, so a random split lets the model use the market of the days it is tested on. In line with this, a random split with market indicators gave an error of 1.6%, whereas on later months the error was 3.0%. Without the indicators, the forward test gives 1.9%, and the indicators were therefore left out of the final model.

3. Data issues and fixes

Issue	Size	Fix
Missing weight	300 (training), 165 (to predict)	Median weight used. The gaps appear random.
Negative weight	292 and 145	Treated as a sign error and replaced by the positive value
Missing market index	374 and 249	Filled with the average of the same day. Not used by the final model.
Cities only in the Nov-Dec loads	8 cities, 12% of loads	Cities described by map position instead of name, so new cities can be handled
Rates 2 to 5 times above or below the usual rate	677 loads (1.4%), about half high and half low	Left out of training

Regarding the unusual rates, none of the load details (distance, weight, equipment, route, date) indicated which loads are affected. Keeping them in training gave a slightly higher error (1.93% against 1.91%), and if Nov-Dec contains a similar share, the final score is likely to be higher than 1.9%. In addition, the inputs were placed on a common scale so that quantities such as miles and pounds are comparable, and the rate per mile was predicted and then multiplied by the distance, since short hauls pay more per mile than long ones.

4. Main drivers of the rate

Distance has the largest effect, as the rate per mile decreases when the haul gets longer. Equipment follows, since Reefer and Flatbed pay more per mile than Dry Van, while weight and route geography have a minor effect. Finally, the market level changes gradually and appears to be lower in Nov-Dec. Since the available data covers eight months, a full yearly cycle could not be observed, and a market component may be worth adding in future work as more data becomes available.

5. Comparison of five models

All five models use the same loads, the same information and the same checks, and a lower error is better.

Model	Error on Sep-Oct loads	Average miss ($)
Straight-line formula (Ridge)	2.26%	52
Many decision trees (Random forest)	2.04%	48
Boosted decision trees	1.90%	45
Small neural network (MLP)	1.92%	45
Support vector regression (SVR), selected	**1.91%**	**45**

Averages over Sep-Oct loads with usual rates.

The straight-line formula has the highest error, which indicates that the link between load details and rate is not linear. The other four are within 0.15 points of each other, so the support vector regression (SVR) was selected for its stable predictions and its handling of new cities rather than for accuracy.

6. Fixed December prediction chart

The chart shows one fixed load: Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb. Because only the date changes through December, the prediction varies between $820 and $837 (average $828). A December market movement is not included, since the available data covers less than one full year. The chart was produced by the provided `score.py`.

